

1. Hello World

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

2. Variables and Data Types

Main.java

```
public class Main {  
    public static void main(String[] args) {  
        int myNumber = 56;           //2147483647  
        long myNum = 214L;  
        float myFloat = 3.14f;       //3.4028235E38  
        double myDouble = 4.5;  
        char myChar = 'A';  
        boolean myBoolean = true;  
        String myString = "Hello, Java";  
  
        System.out.println("Integer: " + myNumber);  
        System.out.println("Long: " + myNum);  
        System.out.println("Float: " + myFloat);  
        System.out.println("Double: " + myDouble);  
        System.out.println("Char: " + myChar);  
        System.out.println("Boolean: " + myBoolean);  
        System.out.println("String: " + myString);  
    }  
}
```

The maximum value an int can hold is 2147483647, and the minimum is -2147483648.

->How to calculate maximum value and minimum value?

int - 4 byte or 32 bit (1 byte = 8 bit)

int is 32 bits = 4 bytes.

One bit is reserved for sign (positive or negative).

Remaining 31 bits are used for the value itself.

Minimum value: $-2^{\{31\}} = -2,147,483,648$

Maximum value: $2^{\{31\}} - 1 = 2,147,483,647$

The maximum value a long can hold is 9223372036854775807, and the minimum is -9223372036854775808.

long (64-bit signed integer):

- Minimum value: $-2^{\{63\}} = -9,223,372,036,854,775,808$
- Maximum value: $2^{\{63\}} - 1 = 9,223,372,036,854,775,807$

3. Arithmetic operators

- Dividing two integers always results in an integer.
- Values like INFINITY and NaN are available **for** floating-point numbers but not **for** integers. As a result, dividing an integer by zero will result in an exception. However, **for** a float or double, Java allows the operation.
- You don't have to import anything; the Math class is in the java.lang package, which is imported by default.

```
public class Main{
    public static void main(String[] args) {
        int a = 12;
        int b = 5;
        int addition = a + b;
        System.out.println(a + " + " + b + " = " + addition);
        int subtraction = a - b;
        System.out.println(a + " - " + b + " = " + subtraction);
        int multiplication = a * b;
        System.out.println(a + " * " + b + " = " +
multiplication);
        int division = a / b;
        System.out.println(a + " / " + b + " = " + division);
        float fdivision = (float)a / b;
        System.out.println(a + " / " + b + " = " + fdivision);
    }
}
```

```

int modulus = a % b;
System.out.println(a + " % " + b + " = " + modulus);

// Increment operator
int increment = a; //12
increment++;
System.out.println(a + " incremented is " + increment);

// Decrement operator
int decrement = a; //12
decrement--;
System.out.println(a + " decremented is " + decrement);

// Float operations
float x = 15.5f;
float y = 4.5f;

// Addition
float floatAddition = x + y;
System.out.println(x + " + " + y + " = " +
floatAddition);

// Subtraction
float floatSubtraction = x - y;
System.out.println(x + " - " + y + " = " +
floatSubtraction);

// Multiplication
float floatMultiplication = x * y;
System.out.println(x + " * " + y + " = " +
floatMultiplication);

// Division
float floatDivision = x / y;
//float floatDivision = (int) (x / y);

```

```
        System.out.println(x + " / " + y + " = " +
floatDivision);

        // Modulus
        float floatModulus = x % y;
        System.out.println(x + " % " + y + " = " +
floatModulus);
    }
}
```

How It Is Calculated

The formula is:

$$a \% b = a - (\text{truncated_quotient} \times b)$$

Important:

The quotient is truncated toward zero.

 Let's Compute $15.5 \% 4.5$

Step 1:

$$15.5 / 4.5 = 3.444\dots$$

Step 2: truncate quotient $\rightarrow 3$

Step 3:

$$15.5 - (3 \times 4.5)$$

= 15.5 - 13.5
= 2.0

So:

15.5f % 4.5f = 2.0

Typecasting :

Type casting in Java is the process of converting a variable from one data type to another. There are two types of casting in Java:

1. Widening Casting (Automatic Type Conversion):

- This is done automatically when you assign a smaller data type to a larger data type.
- Example: **int** to **long**, **float** to **double**.
- No data is lost during this conversion.
- `int i = 100;`
- `long l = i; // automatic widening`
- `System.out.println(l); // Output: 100`

2. Narrowing Casting (Explicit Type Conversion):

- This requires manual conversion because you're assigning a larger data type to a smaller data type.
- Example: **double** to **int**, **float** to **byte**.
- Data may be lost during this conversion, especially if there are decimal points or if the larger type holds values outside the range of the smaller type.

4.String methods

```
public class Main {  
    public static void main(String[] args) {  
        // Create a string  
        String str = "Hello, World!"; //there are other ways  
too.  
  
        // Length of the string  
        int length = str.length();  
        System.out.println("Length of the string: " + length);  
  
        // Convert to uppercase  
        String upperCaseStr = str.toUpperCase();  
        System.out.println("Uppercase string: " + upperCaseStr);  
  
        // Convert to lowercase  
        String lowerCaseStr = str.toLowerCase();  
        System.out.println("Lowercase string: " + lowerCaseStr);  
  
        // Substring from index 7  
        String subStr1 = str.substring(7);  
        System.out.println("Substring from index 7: " +  
subStr1);
```

```

    // Substring from index 0 to 5 (exclusive)
    String subStr2 = str.substring(0, 5);
    System.out.println("Substring from index 0 to 5: " +
subStr2);

    // Replace 'World' with 'Java'
    String replacedStr = str.replace("World", "Java");
    System.out.println("Replaced string: " + replacedStr);

    // Check if the string contains 'World'
    boolean containsWorld = str.contains("World");
    System.out.println("Contains 'World': " +
containsWorld);

    // Check if the string starts with 'Hello'
    boolean startsWithHello = str.startsWith("Hello");
    System.out.println("Starts with 'Hello': " +
startsWithHello);

    // Check if the string ends with '!'
    boolean endsWithExclamation = str.endsWith("!");
    System.out.println("Ends with '!': " +
endsWithExclamation);

    // Trim the string (removing leading and trailing spaces)
    String str1 = "    Hi    ";
    String trimmedStr = str1.trim();
    System.out.println("Trimmed string: '" + trimmedStr +
"'");

    // Split the string by comma
    String[] splitStr = str.split(",");
    System.out.println("Split string by comma: ");
    for (String part : splitStr) {
        System.out.println(part.trim());
    }

```

```
}

// Find the index of 'World'
int indexOfWorld = str.indexOf("World");
System.out.println("Index of 'World': " + indexOfWorld);

// Concatenate with another string
String concatenatedStr = str.concat(" Welcome to
Java!");
System.out.println("Concatenated string: " +
concatenatedStr);
}
}
```

String is not a character array. We cannot get the 5th character using Str[4] – not valid

Str.charAt(4)

5.Array

```
public class Main {  
    public static void main(String[] args) {  
  
        int[] intArray = {10, 20, 30, 40, 50};  
        // int a[] = new int[5];  
  
        // Print the elements of the array  
        System.out.print("Elements of intArray: ");  
        for (int i = 0; i < intArray.length; i++) {  
            System.out.print(intArray[i] + " ");  
        }  
        System.out.println();  
  
        // Find the length of the array  
        int length = intArray.length;  
        System.out.println("Length of intArray: " + length);  
  
        // Accessing elements of the array
```

```

    int firstElement = intArray[0];
    int lastElement = intArray[intArray.length - 1];
    System.out.println("First element: " + firstElement);
    System.out.println("Last element: " + lastElement);

    // Modify an element of the array
    intArray[2] = 35;
    System.out.print("Elements of intArray after
modification: ");
    for (int i = 0; i < intArray.length; i++) {
        System.out.print(intArray[i] + " ");
    }
    System.out.println();

    // Sum of all elements in the array
    int sum = 0;
    for (int i = 0; i < intArray.length; i++) {
        sum += intArray[i];
    }
    System.out.println("Sum of all elements in intArray: " +
sum);

    // Declare and initialize an array of strings
    String[] strArray = {"Java", "is", "fun"};

    // Print the elements of the string array
    System.out.print("Elements of strArray: ");
    for (int i = 0; i < strArray.length; i++) {
        System.out.print(strArray[i] + " ");
    }
    System.out.println();

}
}

```

6. Conditional Statements

```
public class Conditionals {  
    public static void main(String[] args) {  
        int x = 10;  
        int y = 20;  
  
        if (x > y) {  
            System.out.println("x is greater than y");  
        } else if (x < y) {  
            System.out.println("x is less than y");  
        } else {  
            System.out.println("x is equal to y");  
        }  
    }  
}
```

7. Loops

```
public class Loops {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println("i = " + i);  
        }  
    }  
}
```

8. for and enhanced for loop

```
public class Main {  
    public static void main(String[] args) {  
  
        int[] intArray = {10, 20, 30, 40, 50};  
  
        System.out.print("Elements of intArray using for loop: ");  
        for (int i = 0; i < intArray.length; i++) {  
            System.out.print(intArray[i] + " ");  
        }  
    }  
}
```

```

        System.out.println();

        System.out.print("Elements of intArray using for-each loop:
");
        //enhanced for loop
        for (int element : intArray) {
            System.out.print(element + " ");
        }
    }
}

```

9. while loop

```

public class Main {
    public static void main(String[] args) {
        int number = 1;
        while (number <= 5) {
            System.out.println(number);
            number++;
        }
    }
}

```

10. do while loop

```

public class Main {
    public static void main(String[] args) {
        int number = 1;
        do {
            System.out.println(number);
            number++;
        } while (number <= 5);
    }
}

```

11. Switch

```
public class Main {  
    public static void main(String[] args) {  
        int day = 1;  
        String dayName;  
        switch (day) {  
            case 1:  
                dayName = "Monday";  
                break;  
            case 2:  
                dayName = "Tuesday";  
                break;  
            case 3:  
                dayName = "Wednesday";  
                break;  
            default:  
                dayName = "Invalid day";  
                break;  
        }  
        System.out.println("Day " + day + " is " + dayName);  
    }  
}
```

12. Functions

Static methods are usually used for utility methods that do not require any instance specific data.

```
public class Functions {  
    // Function to add two numbers
```

```

    public static int add(int a, int b) {
        return a + b;
    }

    public static void main(String[] args) {
        int x = 5;
        int y = 10;
        int sum = add(x, y);
        System.out.println("Sum: " + sum);
    }
}

```

13. Reading Input from the User

13.1 Using console (not used commonly)

```

import java.io.Console;

public class Main {
    public static void main(String[] args) {
        Console console = System.console();

        String name = console.readLine("Enter your name: ");
        char[] password = console.readPassword("Enter your password:
");

        String passwordString = new String(password);
    }
}

```

```

        System.out.println("Hello, " + name + "!");
        System.out.println("Your password is: " + passwordString);
    }
}

```

13.2 Using Scanner Class

```

import java.util.Scanner;

public class UserInput {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        System.out.println("Hello, " + name + ". You are " + age + "
years old.");
    }
}

```

13.3 Difference between next(),nextLine(),nextInt

```

import java.util.Scanner;

public class Main{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
    }
}

```

```

    // Reading a single word
    System.out.print("Enter a word: ");
    String word = scanner.next();
    System.out.println("You entered: " + word);

    // Reading a line of text
    System.out.print("Enter a sentence: ");
    scanner.nextLine(); // Consume newline left by previous next()
call
    String sentence = scanner.nextLine();
    System.out.println("You entered: " + sentence);

    // Reading an integer
    System.out.print("Enter an integer: ");
    int number = scanner.nextInt();
    System.out.println("You entered: " + number);
}
}

```

14. To check odd or even number

```

import java.util.Scanner;

public class OddOrEven {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = scanner.nextInt();

        if (num % 2 == 0) {
            System.out.println(num + " is even.");
        } else {
            System.out.println(num + " is odd.");
        }
    }
}

```

```
        scanner.close();
    }
}
```

15. Factorial

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = scanner.nextInt();
        int factorial = 1;

        for (int i = 1; i <= num; i++) {
            factorial *= i;
        }

        System.out.println("Factorial of " + num + " is " +
factorial);

        scanner.close();
    }
}
```

16. Simple Interest Calculator

```
import java.util.Scanner;

public class SimpleInterest {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```

        System.out.print("Enter principal amount: ");
        double principal = scanner.nextDouble();

        System.out.print("Enter rate of interest: ");
        double rate = scanner.nextDouble();

        System.out.print("Enter time (in years): ");
        double time = scanner.nextDouble();

        double simpleInterest = (principal * rate * time) / 100;

        System.out.println("Simple Interest: " + simpleInterest);

        scanner.close();
    }
}

```

17. Sum of array elements

```

import java.util.Scanner;

public class SumArray {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of elements in the array:
");

        int n = scanner.nextInt();
        int[] array = new int[n];

        System.out.println("Enter the elements of the array:");
        for (int i = 0; i < n; i++) {
            array[i] = scanner.nextInt();
        }

        int sum = 0;
        for (int i = 0; i < n; i++) {
            sum += array[i];
        }
    }
}

```

```

    }

    System.out.println("Sum of array elements: " + sum);

    scanner.close();
}
}

```

18. Palindrome checker

```

import java.util.Scanner;

public class PalindromeChecker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = scanner.nextLine();

        String reversedStr = new
        StringBuilder(str).reverse().toString();

        if (str.equals(reversedStr)) {
            System.out.println(str + " is a palindrome.");
        } else {
            System.out.println(str + " is not a palindrome.");
        }

        scanner.close();
    }
}

```

19. Fibonacci series

```

import java.util.Scanner;

public class FibonacciSeries {

```

```

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter the number of terms: ");
    int n = scanner.nextInt();

    int a = 0, b = 1, next;

    System.out.print("Fibonacci series: " + a + " " + b);

    for (int i = 2; i < n; i++) {
        next = a + b;
        System.out.print(" " + next);
        a = b;
        b = next;
    }

    scanner.close();
}
}

```

20. Basic Object-Oriented Programming

```

class Dog {
    // Properties
    String name;
    int age;

    // Constructor
    Dog(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Method
    void bark() {

```

```

        System.out.println(name + " says: Woof!");
    }
}

public class OOPExample {
    public static void main(String[] args) {
        Dog myDog = new Dog("Buddy", 3);
        System.out.println("My dog's name is " + myDog.name + " and he
is " + myDog.age + " years old.");
        myDog.bark();
    }
}

```

```

Class A:
method1(){
    S.out();
    S.out();
}

```

```

....
A a = new A()
a.method1();
...

```